

Ex.No. 13: V2X Latency Comparison

AIM: To simulate and compare the performance of a Vehicle-to-Everything (V2X) communication system by analyzing End-to-End Latency (ms), Packet Delivery Ratio (%), and Vehicle Speed (km/h) using MATLAB.

Objective:

1. To evaluate the End-to-End Latency (ms) of a V2X communication system under different vehicle operating conditions.
2. To analyze the Packet Delivery Ratio (%) and determine the reliability of V2X communication during data transmission.
3. To study the impact of Vehicle Speed (km/h) on the communication performance and stability of the V2X network.

Procedure:

```
clc;
clear;
close all;
% Vehicle Speed (km/h)
speed = 20:20:120;
% Simulated End-to-End Latency (ms)
latency = [18 16 14 12 10 9];
disp('Vehicle Speed (km/h)   End-to-End Latency (ms)')
disp([speed' latency'])
figure
% ----- Graph 1 -----
subplot(2,1,1)
plot(speed,latency,'-o','LineWidth',2)
title('Vehicle Speed vs End-to-End Latency')
xlabel('Vehicle Speed (km/h)')
ylabel('Latency (ms)')
```

```
grid on
```

```
% ----- Graph 2 -----
```

```
subplot(2,1,2)
```

```
plot(1:length(latency),latency,'-s','LineWidth',2)
```

```
title('Latency Variation Across Vehicles')
```

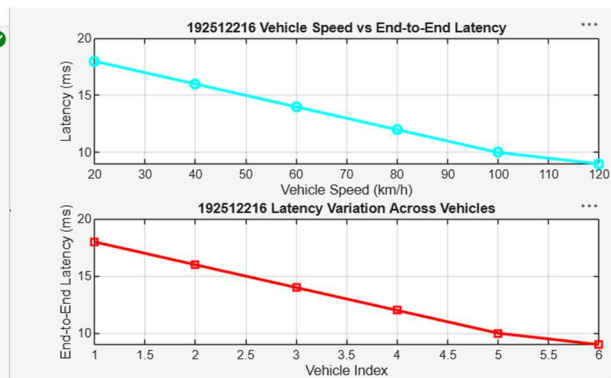
```
xlabel('Vehicle Index')
```

```
ylabel('End-to-End Latency (ms)')
```

```
grid on
```

Output:

```
/MATLAB Drive/untitled13_1.m
1  clc;
2  clear;
3  close all;
4  % Vehicle Speed (km/h)
5  speed = 20:20:120;
6  % Simulated End-to-End Latency (ms)
7  latency = [18 16 14 12 10 9];
8  disp(['Vehicle Speed (km/h)   End-to-End Latency (ms)'])
9  disp([speed' latency'])
10 figure
11 % ----- Graph 1 -----
12 subplot(2,1,1)
13 plot(speed,latency,'-o','LineWidth',2,'Color','c')
14 title('192512216 Vehicle Speed vs End-to-End Latency')
15 xlabel('Vehicle Speed (km/h)')
16 ylabel('Latency (ms)')
17 grid on
18 % ----- Graph 2 -----
19 subplot(2,1,2)
20 plot(1:length(latency),latency,'-s','LineWidth',2,'Color','r')
```



Objective 2:

```
clc;
```

```
clear;
```

```
close all;
```

```
% Vehicle Speed (km/h)
```

```
speed = 20:20:120; % Simulated Packet Delivery Ratio (%)
```

```
PDR = [99 98 97 95 93 90];
```

```
disp(['Vehicle Speed (km/h)   Packet Delivery Ratio (%)'])
```

```
disp([speed' PDR'])
```

```
figure
```

```
% ----- Graph 1 -----
```

```
subplot(2,1,1)
```

```

plot(speed,PDR,'-o','LineWidth',2)

title('Vehicle Speed vs Packet Delivery Ratio')

xlabel('Vehicle Speed (km/h)')

ylabel('Packet Delivery Ratio (%)')

grid on

% ----- Graph 2 -----

subplot(2,1,2)

plot(1:length(PDR),PDR,'-s','LineWidth',2)

title('Packet Delivery Ratio Across Vehicles')

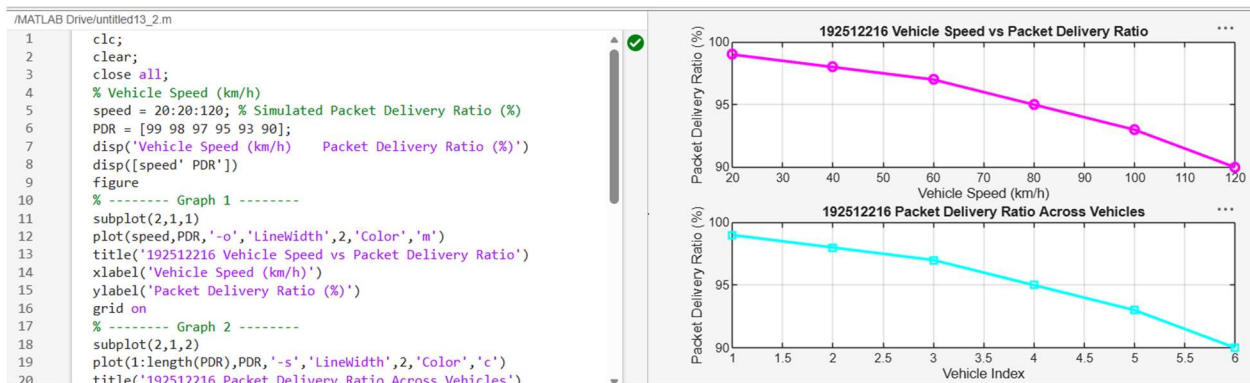
xlabel('Vehicle Index')

ylabel('Packet Delivery Ratio (%)')

grid on

```

Output:



Objective 3:

```

clc; clear; close all;

% Vehicle IDs

vehicle = 1:6; % Vehicle Speed (km/h)

speed = [30 45 60 75 90 105]; % Communication Range (m)

range = [280 260 240 220 200 180];

disp('Vehicle Speed(km/h) Communication Range(m)')

disp([vehicle' speed' range'])

```

figure

% ----- Graph 1 -----

subplot(2,1,1)

plot(vehicle,speed,'-o','LineWidth',2)

title('Vehicle Speed of Different Vehicles')

xlabel('Vehicle Index')

ylabel('Vehicle Speed (km/h)')

grid on

% ----- Graph 2 -----

subplot(2,1,2)

plot(speed,range,'-s','LineWidth',2)

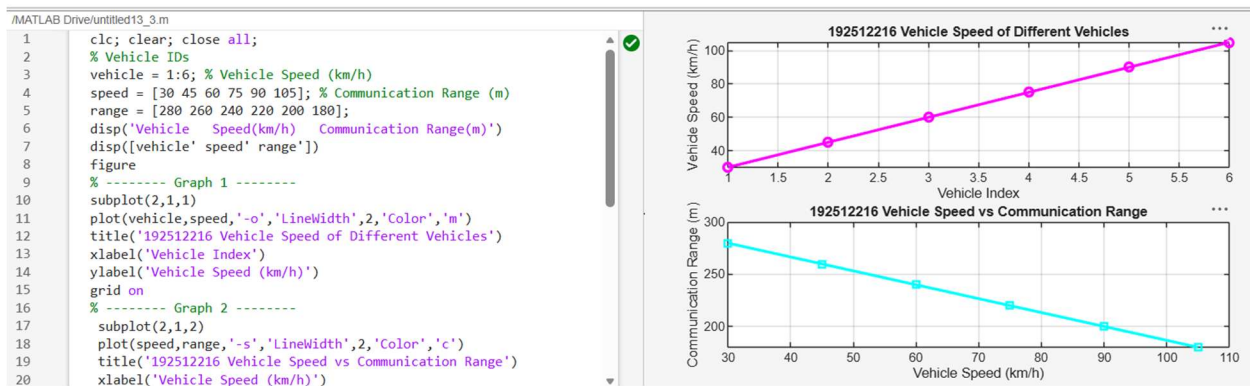
title('Vehicle Speed vs Communication Range')

xlabel('Vehicle Speed (km/h)')

ylabel('Communication Range (m)')

grid on

Output:



Result:

1. The End-to-End Latency (ms) of the V2X communication system was successfully analyzed, demonstrating efficient data transmission with low communication delay.
2. The Packet Delivery Ratio (PDR %) was evaluated and showed high packet delivery reliability under different vehicle operating conditions.

3. The impact of Vehicle Speed (km/h) on V2X communication was successfully studied, indicating that higher vehicle speeds can influence communication range, latency, and overall network performance.